Revision | **Covers** 1-1 → 1-3 | **File** 5 / 7

REV

Chapter 1 revision pack

حزمة المراجعة — شجرة التصنيف، الحالات الحدّية، الأخطاء الشائعة، وخطة A+

1 · Classify any variable in three questions

Q1 **Category or number with arithmetic meaning?**

Category → qualitative (nominal or ordinal). Number → quantitative (interval or ratio). A number used only as a label (ID, zip code, jersey number) counts as a category.

Q2 **If a category: does it have a natural order?**

No → nominal. Yes → ordinal.

Q3 **If a number: counted or measured?**

Counted → discrete; measured → continuous. Then **does zero mean "none"?**

Yes → ratio. No → interval.

2 · Borderline cases table

VARIABLE	TYPE	LEVEL	WHY
Student ID 10234	qualitative	nominal	the number is only a label
Letter grade A, B, C	qualitative	ordinal	ordered, but the gaps are not equal
Exam mark out of 100	quantitative, continuous	ratio	0 marks = none (Dr. Ahmed's list)
GPA	quantitative, continuous	ratio	both decks
Temperature °C / °F	quantitative, continuous	interval	0 °C ≠ no heat
Temperature in Kelvin	quantitative, continuous	ratio	0 K = absolute zero
Calendar year, date, clock time	quantitative	interval	year 0 is not "no time"

VARIABLE	TYPE	LEVEL	WHY
Time taken (45 minutes)	quantitative, continuous	ratio	a duration: 0 min = none
Money earned per week	quantitative, **discrete**	ratio	Bluman: counted in the smallest unit
Likert item (agree … disagree)	qualitative	ordinal	even when scored 1–5
IQ, SAT score	quantitative	interval	no true zero
Number of steps per day	quantitative, discrete	ratio	counted; 0 steps = none

3 · Sampling method, recognised from the wording

THE SCENARIO SAYS…	METHOD
"randomly generated numbers", "names drawn from a box", "random number table"	**simple random**
"every 10th customer", "every 5th name on the list"	**systematic**
"grouped by … then a number *from each group* randomly selected"	**stratified**
"randomly chose 5 schools / 2 buildings and surveyed *everyone* in them"	**cluster**
"the first 10 who arrived", "people leaving the mall"	**convenience**
"those who chose to respond", "an online poll anyone can answer"	**volunteer**

4 · Ten traps

1. Calling a numeric ID quantitative.
2. Calling a sample percentage ("55% of those surveyed") inferential: it is descriptive.
3. Putting year or °C at the ratio level.
4. Treating letter grades as interval because they look like scores.
5. Rounding k **up** in systematic sampling.
6. Keeping a repeated or out-of-range number from Table D.
7. Reading Table D in groups that don't match the number of digits in N.
8. Swapping stratified and cluster: is it *some from every group*, or *all from some groups*?
9. Calling convenience or volunteer samples random.
10. Calling a defective instrument "sampling error": it is **nonsampling** error.

5 · R for Lab 1 (outline: Get RStudio, Get started with R, R Project, R Markdown)

TASK	R
store data / look at a data set	`x <- c(150, 165, 172, 180, 190)` `head(df)` `str(df)`

TASK	R
variable type in R (qualitative is a *factor*, quantitative is *numeric*)	`class(df$gender)` `df$grade <- factor(df$grade, levels = c("C","B","A"), ordered = TRUE)`
simple random sample of n = 10 from 1 to 4998	`sample(1:4998, 10)` (without replacement by default)
systematic sample, k = 14, start 6	`seq(6, 86, by = 14)`

6 · A+ plan for Chapter 1

1 **Learn the definitions as sentences**
: statistics, data, datum, variable, population, sample, census, survey, descriptive, inferential, biased sample. Each is a one-mark CLO1-A question.

2 **Drill the borderline table**
until you can justify every row in one line, because classification questions are built on exactly those rows.

3 **Do one SRS from Table D and one systematic sample by hand**
without notes. These are the only calculations in the chapter.

4 **Textbook page 34, Q 6, 22, 39, 42**
(assigned in the outline), then the question bank.

خطة A+ بالعربي

(1) احفظ التعاريف كجمل كاملة. (2) تدرّب على جدول الحالات الحدّية حتى تبرّر كل صف بسطر واحد. (3) طبّق مرة واحدة **عينة عشوائية بجدول الأرقام** ومرة **عينة منتظمة** بيدك دون النظر للحل. (4) حل تمارين الكتاب صفحة 34 (6، 22، 39، 42) ثم بنك الأسئلة.

Chapter 1 study set: 7 files. One merged source built from Dr. Amani Al-Rubkhi's Fall 2026 notes (course coordinator) and Dr. Ahmed Al-Hadhrami's slide pack, aligned to the SQU STAT 2101 course outline (Fall 2026). Textbook: Bluman, *Elementary Statistics*, 10th ed.

00 · Exam brief · 01 · Lesson 1-1: Descriptive and Inferential Statistics · 02 · Lesson 1-2: Variables and Types of Data · 03 · Lesson 1-3: Data Collection and Sampling · **04 · Revision pack** · 05 · Exam-style questions · 06 · Full worked answers